

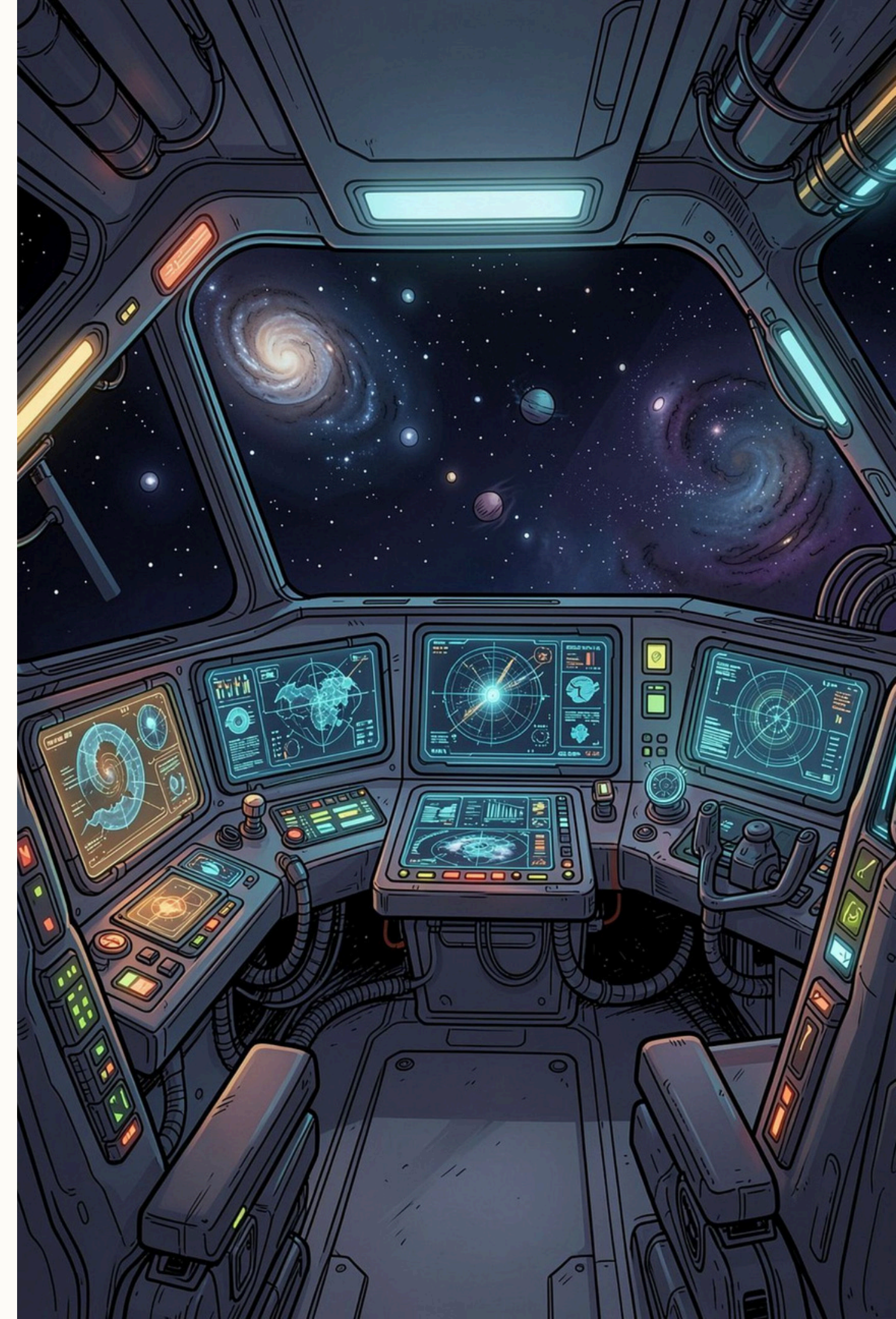
Spacecraft Life-Support Simulator

Design and Simulation of an Arduino-Based Multi-Parameter Environmental Monitoring and Emergency Response System

TEMPERATURE · GAS · LIGHT · EMERGENCY

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"A compact spacecraft cabin is a controlled physical environment. Continuous sensing, decision-making and automatic response are therefore essential to crew safety."



Abstract

An Arduino-based educational prototype that monitors spacecraft cabin conditions and responds automatically when unsafe levels are detected.

- **What it does:** tracks temperature, gas, and light levels.
- **How it works:** TMP36, MQ-2, and LDR sensors feed an LCD, LEDs, buzzer, and SG90 servo.
- **Engineering principle:** sense → decide → communicate → act.

This project shows how a low-cost microcontroller can combine sensing and response logic in a simple model of spacecraft life-support systems.

Abstract & Scientific Context

What It Does

An Arduino-based educational prototype that continuously monitors critical cabin parameters and automatically reacts when unsafe conditions are detected. Integrates TMP36 temperature, MQ-2 gas, and LDR light sensors with LCD, LEDs, buzzer, and SG90 servo.

Core Research Question

How can a low-cost microcontroller integrate multiple environmental sensors and produce a coordinated automatic safety response for a simulated spacecraft cabin?

The engineering principle abstracted: **sense** → **decide** → **communicate** → **act** — demonstrated as a low-cost educational model of real spacecraft life-support systems.

ARDUINO UNO

TMP36

MQ-2

LDR

TINKERCAD

Project Objectives

The project addresses seven interconnected goals spanning sensing, decision logic, output, and evaluation.

1	Multi-Parameter Architecture Design a compact spacecraft cabin monitoring architecture integrating temperature, gas, and light sensing.
2	Analogue Sensing Measure temperature, gas-sensor output, and ambient-light level as independent analogue inputs via Arduino ADC.
3	Threshold Definition Define safe operating bands and abnormal-condition thresholds for each monitored parameter.
4	Real-Time Display Display measurements and system status live on a 16×2 LCD.
5	Warning Indicators Provide visual (red/green LEDs) and audible (buzzer) warning indicators for abnormal conditions.
6	Automatic Mechanical Response Demonstrate an automatic emergency response using an SG90 servo representing a hatch or isolation door.
7	Evaluation & Future Scope Evaluate advantages, limitations, applications, and propose future improvements for research extension.

System Architecture & Components

The architecture is organised around the Arduino UNO as the decision-making unit. Each sensor produces an electrical signal proportional to a physical condition, read through analogue inputs A0–A2.



Physical environment → Sensors → Arduino ADC → Threshold/decision logic → LCD + LEDs + buzzer → Servo emergency mechanism.

Key Components

- **Arduino UNO R3** — Central controller and ADC (₹550)
- **TMP36** — Cabin temperature measurement (₹120)
- **MQ-2** — Gas/smoke/combustible-gas indication (₹130)
- **LDR + Resistor** — Ambient light measurement (₹15)
- **16×2 LCD** — Live values and system state (₹100)
- **SG90 Servo** — Emergency hatch/mechanism (₹100)
- **Buzzer + LEDs** — Audible and visual alarms (₹30)

Budget Summary

Estimated hardware subtotal: approximately **₹1,255**. With miscellaneous items, enclosure, labels, and contingency, a practical student prototype budget is approximately **₹1,400–₹1,600**. Tinkercad simulation serves as a zero-cost modelling stage. Prices are indicative and vary by seller, quality, and location.

Circuit Connections & Sensor Logic

The circuit assigns three analogue inputs and several digital outputs. Sensors connect to A0–A2; actuators and display connect to digital pins D2–D12.

Device	Arduino Pin	Power	Purpose
TMP36	A0	5 V, GND	Analogue temperature voltage
MQ-2 Module	A1	5 V, GND	Analogue gas-sensor level
LDR Divider	A2	5 V, GND	Analogue light level
LCD (RS, E, D4–D7)	D12, D11, D5–D2	5 V, GND	4-bit data bus
Buzzer	D7	GND	Audible warning
Red LED	D8	GND	Alarm indicator
Green LED	D9	GND	Safe indicator
Servo Signal	D10	5 V, GND	Emergency mechanism
LCD Contrast	10 kΩ Pot	5 V, GND	Contrast control

TMP36

Analogue voltage changes approximately linearly with temperature. Above 30 °C = high-temperature event.

MQ-2 & LDR

MQ-2 metal-oxide sensor output changes with gas concentration; analogue reading used as gas-risk proxy (threshold > 400). LDR photoresistor with resistor forms a voltage divider for light-level estimation (threshold < 300 = low light).

Working Principle & Control Algorithm

The simulator operates as a closed-loop monitoring system. The Arduino repeatedly samples three analogue channels, converts readings to meaningful quantities, compares them against safety thresholds, and updates the LCD and actuators accordingly.

Control Sequence

1 Initialise

Set up LCD, buzzer, LEDs, and servo in setup().

2 Sample

Read temperature from A0, gas from A1, light from A2.
Convert TMP36 voltage to approximate °C.

3 Decide

Compare readings against thresholds: temp > 30 °C, gas > 400, light < 300.

4 Respond

All safe → green LED + SYSTEM NORMAL. One unsafe → red LED + buzzer + warning. All three unsafe → EMERGENCY: ALL FAIL + servo opens to 90°.

📄 **Engineering Note:** Thresholds are demonstration values, not spacecraft safety limits. A real life-support system would require calibrated sensors, redundancy, fault detection, validated atmospheric models, watchdogs, and extensive qualification testing.

Threshold Model & Operating States

Five principal operating states are defined by the combination of temperature, gas, and light readings. The system responds with graded warnings, escalating to a full emergency response when all three parameters are simultaneously unsafe.

Condition	Temperature	Gas	Light	Response	Servo
All Normal	20–30 °C	< 400	> 300	Green LED; SYSTEM NORMAL	Closed
High Temperature	> 30 °C	< 400	> 300	Red LED + Buzzer; WARNING: HIGH TEMP	Normal
Gas Detected	20–30 °C	> 400	> 300	Red LED + Buzzer; WARNING: GAS DETECTED	Normal
Low Light	20–30 °C	< 400	< 300	Red LED + Buzzer; WARNING: LOW LIGHT	Normal
All Fail (Emergency)	> 30 °C	> 400	< 300	EMERGENCY: ALL FAIL; Buzzer + Servo	Open (90°)



Normal

Green LED ON. LCD: SYSTEM NORMAL. Servo closed.



Single Fault

Red LED + Buzzer ON. LCD shows specific warning. Servo normal.



All Fail

Full emergency: buzzer + red LED + servo opens to 90°. LCD: EMERGENCY: ALL FAIL.

Arduino Code & Tinkercad Simulation

Complete spacecraft life-support simulator logic for TMP36 temperature, MQ-2 gas detection, and LDR light monitoring, with LCD feedback, LED alerts, buzzer control, and servo-based emergency response.

Complete Arduino Code

```
#include <LiquidCrystal.h>
#include <Servo.h>

const int tempPin = A0;      // TMP36
const int gasPin = A1;       // MQ-2
const int lightPin = A2;     // LDR

const int greenLed = 7;
const int redLed = 8;
const int buzzer = 9;
const int servoPin = 10;

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
Servo hatchServo;

void setup() {
  pinMode(greenLed, OUTPUT);
  pinMode(redLed, OUTPUT);
  pinMode(buzzer, OUTPUT);

  digitalWrite(greenLed, LOW);
  digitalWrite(redLed, LOW);
  digitalWrite(buzzer, LOW);

  hatchServo.attach(servoPin);
  hatchServo.write(0); // hatch closed

  lcd.begin(16, 2);
  lcd.clear();
  lcd.print("SYSTEM STARTING");
  delay(1500);
}

void loop() {
  int tempRaw = analogRead(tempPin);
  int gasValue = analogRead(gasPin);
  int lightValue = analogRead(lightPin);

  float voltage = tempRaw * (5.0 / 1023.0);
  float tempC = (voltage - 0.5) * 100.0; // TMP36 conversion

  bool tempHigh = (tempC > 30.0);
  bool gasDetect = (gasValue > 400);
  bool lightLow = (lightValue < 300);

  lcd.clear();
```

Tinkercad Simulation States

- State 1 — All Normal: green LED on, LCD shows SYSTEM NORMAL, hatch closed
- State 2 — High Temp: red LED and buzzer on, LCD shows WARNING: HIGH TEMP
- State 3 — Gas Detected: red LED and buzzer on, LCD shows WARNING: GAS DETECTED
- State 4 — Low Light: red LED and buzzer on, LCD shows WARNING: LOW LIGHT
- State 5 — All Fail: emergency mode, servo opens to 90°, LCD shows EMERGENCY

Advantages, Limitations & Applications

✓ Advantages

- Low-cost architecture (~₹1,400–₹1,600) using commonly available educational components
- Multi-parameter monitoring combines several environmental indicators in one controller
- Real-time LCD feedback with red/green LEDs and buzzer for fast warning channels
- Servo-based emergency action demonstrates closed-loop automation
- Tinkercad simulation allows safe testing without hazardous conditions
- Modular design permits addition of further sensors

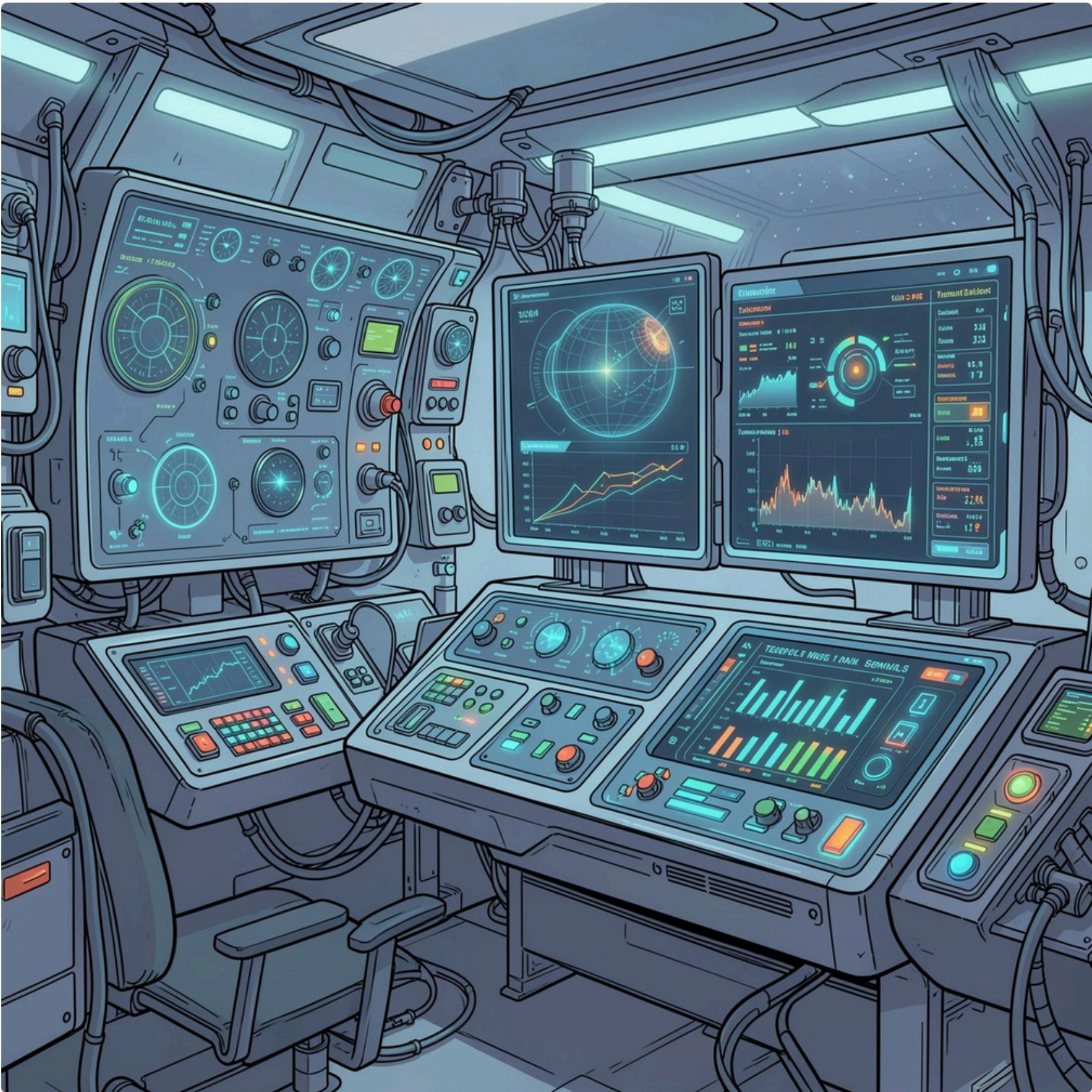
⚠ Limitations

- MQ-2 is not a selective spacecraft gas analyser; requires calibration
- TMP36 and LDR lack precision, redundancy, and qualification for real spacecraft systems
- Single-sensor architecture creates a potential single point of failure
- Fixed thresholds can generate false alarms
- Tinkercad does not reproduce real spacecraft atmospheric dynamics
- Power integrity, sensor warm-up, noise, and ADC resolution not fully addressed

🚀 Applications

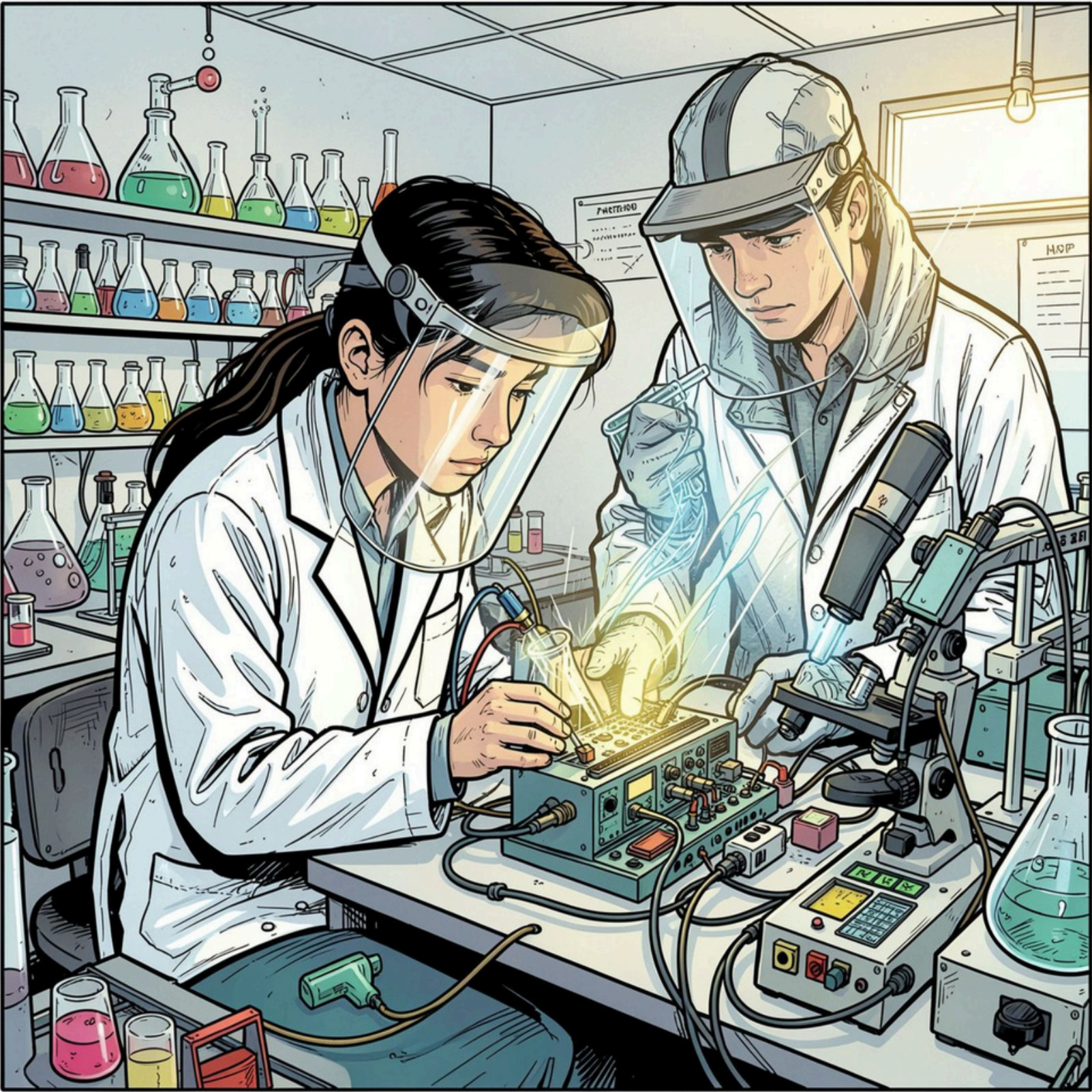
- Space-science and National Space Day exhibitions
- Physics laboratory demonstrations of sensors and analogue-to-digital conversion
- Arduino and embedded-system education
- Environmental monitoring prototypes for controlled rooms
- STEM demonstrations of automation, telemetry, and emergency response
- Concept validation for cabin-monitoring research

Future Scope, Safety & Conclusion



Proposed Future Enhancements

- **Atmospheric sensing:** Add CO₂, O₂, pressure, humidity, and smoke sensors
- **Sensor fusion:** Multiple sensors per parameter to identify drift or failure
- **Data logging:** Store time-stamped measurements for trend analysis
- **Telemetry:** Send cabin status to a ground-station dashboard
- **Adaptive thresholds:** Calibrated, context-dependent alarm bands



Safety & Ethics

Demonstrate using safe simulation values only — never intentionally create hazardous gas environments. The MQ-2 response in Tinkercad is a teaching abstraction, not a certified gas detector. Any physical prototype must be tested in a controlled laboratory with appropriate supervision.

Conclusion